

Lab 1

Ex 1

$$(D_1) = (1, 0, 1, 1, 0)$$

$$(D_2) = (0, 1, 1, 1, 0)$$

$$(D_3) = (1, 0, 0, 1, 1)$$

Ex 2

$$t.f(\text{cat}, D_1) = \frac{c(\text{cat}, D_1)}{\sum_{\text{cat}} c(\text{cat}, d_i)} = \frac{1}{3}$$

$$t.f(\text{cat}, D_2) = \frac{1}{3}$$

$$t.f(\text{fish}, D_1) = \frac{1}{3}$$

Ex 3.

$$id.f(\text{cat}) = \log \frac{3}{d.f(\text{cat})} = \log \frac{3}{2}$$

$$id.f(\text{log}) = \log \frac{3}{1}$$

$$id.f(\text{cats}) = \log \frac{3}{2}$$

$$id.f(\text{fish}) = \log 1$$

$$id.f(\text{likes}) = \log 3$$

$id.f(\text{fish}) = 0$ thấp nhất $\rightarrow$ id.f càng cao thì term đó càng hiếm

Ex 4.

$$i.f - id.f(\text{cat}) = \frac{1}{3} \log \frac{3}{2}$$

$$i.f - id.f(\text{cats}, D_1) = \frac{1}{3} \log \frac{3}{2}$$

$$i.f - id.f(\text{fish}, D_1) = \frac{1}{3} \cdot 0 = 0$$

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Ex 5

$$x = [1, 1, 1]$$

$$y = [1, 1, 0]$$

$$\cos(x, y) = \frac{x^T \cdot y}{\|x\|_2 \cdot \|y\|_2} = \frac{2}{\sqrt{6}}$$

Ex 6

1. D_1 cao nhất

2. D_2 thấp nhất

3. "Medical" "image" có 1 DF thấp.